MIRROR: A Hierarchical Benchmark for Metacognitive Calibration in Large Language Models

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Abstract

We introduce MIRROR (Metacognitive Integrity and Recursive Reasoning in Operationalized Response), a benchmark comprising eight experiments across four metacognitive levels that evaluates whether large language models can use self-knowledge to make better decisions. We evaluate 16 models from 8 labs across approximately 250,000 evaluation instances using five independent behavioral measurement channels. We find two phenomena with direct implications for agentic deployment: (1) compositional self-prediction fails universally—the Compositional Calibration Error ranges from 0.500 to 0.943 across 15 models, and the Metacognitive Convergence Index is 0.000, and (2) models possess accurate domain-specific self-knowledge but systematically fail to translate it into appropriate agentic action-selection—external metacognitive control reduces the Confident Failure Rate from 0.562 to 0.214, a 62% reduction. Providing models with their own calibration scores produces no significant improvement ($p > 0.05$); only architectural constraint is effective. These findings challenge the assumption underlying current agentic architectures that accurate self-knowledge enables self-correction, and suggest that external metacognitive scaffolding—not improved self-knowledge—is the path to safer autonomous AI systems. MIRROR and all data are publicly available.

1 Introduction

The deployment of large language models (LLMs) as autonomous agents presupposes a form of metacognitive competence: the model must not only know *what* it knows, but must translate that awareness into appropriate action—proceeding when confident, escalating to tools or human review when uncertain. This assumption underlies agentic architectures built on self-reflection loops, confidence-based routing, and Constitutional AI self-monitoring [Perez et al., 2022]. It is foundational to the enterprise of agentic AI safety. It has never been systematically tested.

Prior work has studied LLM calibration [Kadavath et al., 2022, Kapoor et al., 2024], uncertainty estimation [Lin et al., 2024], and situational awareness [Laine et al., 2024]. But these address different questions: calibration measures whether confidence matches accuracy; situational awareness measures whether models know *what they are*. Nobody has systematically tested whether models can *use* their self-knowledge to make better decisions—whether the bridge from metacognitive *monitoring* to metacognitive *control* [Nelson and Narens, 1990] is intact.

We introduce MIRROR, a hierarchical benchmark spanning four metacognitive levels (atomic self-knowledge, cross-domain transfer, compositional prediction, and adaptive self-regulation), eight experiments, and five independent behavioral measurement channels. We evaluate 16 models from 8 labs across approximately 250,000 evaluation instances.[1]

Three findings emerge. First, compositional self-prediction fails universally: the Compositional Calibration Error (CCE) ranges from 0.500 to 0.943 across 15 models, and the Metacognitive Convergence Index (MCI) is 0.000—models that calibrate accurately in individual domains completely

[1]Computed as: Exp1 (5,000 $\times$ 5 channels $\times$ 16 models) + Exp4 (320 $\times$ 2 conditions $\times$ 15 models) + Exp9 (597 $\times$ 4 conditions $\times$ 3 paradigms $\times$ 10 models) + Experiments 2, 3, 5, 6, 8 ($\sim$30,000 total). See Section 3 for per-experiment details.

Table 1: Comparison of MIRROR with prior benchmarks for LLM self-knowledge and metacognition.

Feature	SAD (Laine 2024)	Barkan et al. (2025)	Kadavath et al. (2022)	Steyvers et al. (2025)	MIRROR (Ours)
Venue	NeurIPS D&B '24	NeurIPS WS '25	arXiv	arXiv	NeurIPS D&B '26
Focus	Situational awareness	Self-capability prediction	Calibration (P(IK))	Metacognition + uncertainty	Metacognitive cal. → agentic action
Models	16	∼15	∼10	2 families	16
Task categories	7	3	—	6 datasets	8 experiments
Total items	13K questions	∼1K tasks	—	12K+ items	∼250K records
Metacognitive levels	1 (awareness)	1 (confidence)	1 (calibration)	1 (calibration)	4 (L0–L3)
Behavioral channels	1 (MC/free)	1 (verbal)	1 (verbal)	1 (verbal)	5 (wager, opt-out, tool, diff., natural)
Agentic connection	No	Yes (SWE-bench)	No	No	Yes (4 cond., 3 par.)
Headline finding	Partial sit. awareness	LLMs over-confident	*Mostly* calibrated	Task-specific metacognition	Knowing-doing gap (62% CFR reduction)
Public dataset	Yes	No	No	No	Yes

fail to compose those calibrations for multi-domain tasks. Second, the knowing-doing gap: models possess accurate self-knowledge but do not act on it. External constraint reduces the Confident Failure Rate (CFR) from 0.562 to 0.214 (62% reduction); providing models with their own calibration scores produces no significant improvement. Third, self-knowledge is domain-atomic: the Transfer Influence Index ranges from 0.019 to 0.175 across 11 models, indicating that calibration does not meaningfully cross domain boundaries.

Contributions.

- The MIRROR benchmark: 8 experiments across 4 metacognitive levels, 5 behavioral channels, 5,000 domain-stratified questions, and 597 composite agentic tasks, publicly released with all protocols and analysis code.
- Two empirical findings—universal compositional failure (MCI = 0.000) and the knowing-doing gap (62% CFR reduction under external control)—that challenge assumptions about LLM self-knowledge.
- Quantification of the knowing-doing gap's cost: in a 1,000-task agentic deployment, ∼562 tasks fail silently without external constraint versus ∼214 with constraint.
- Evidence that external metacognitive scaffolding, not better self-knowledge, is the path to safer agentic systems.

Table 1 positions MIRROR relative to prior work.

2 Related Work

LLM Calibration. Kadavath et al. [2022] showed that LLMs can predict their own answer correctness above chance on knowledge-intensive tasks. Kapoor et al. [2024] demonstrated that fine-tuning is needed for reliable calibration. Steyvers et al. [2025] found that metacognitive ability is task-specific, not general—a finding consistent with MIRROR's domain-level analysis. MIRROR differs from all prior calibration work in testing whether calibration *transfers to action-selection*: we measure the gap between knowing and doing, not the accuracy of knowing alone.

LLM Self-Knowledge and Situational Awareness. SAD [Laine et al., 2024] measures whether LLMs know what they are—identity, lifecycle stage, deployment context. Barkan et al. [2025] tested capability prediction on SWE-bench tasks, finding systematic overconfidence. Xiong et al. [2024] found that LLMs are often overconfident when expressing numerical probability estimates. MIRROR is complementary: SAD diagnoses awareness; MIRROR diagnoses the gap between awareness and behavior.

Agentic Evaluation. AgentBench [Liu et al., 2024], OSWorld [Xie et al., 2024], and GAIA [Mialon et al., 2023b] measure task success rate in agentic settings. These benchmarks evaluate execution quality conditional on the decision to act. Concurrent work on agentic confidence calibration [Anonymous,

2026a] measures calibration during multi-step trajectories using trajectory-level features. MIRROR differs in measuring metacognitive calibration *before* deployment and testing whether pre-deployment assessment predicts agentic outcomes. More broadly, MIRROR inverts the standard framing: we measure the *decision* itself, treating task execution quality as a downstream variable and calibration-informed decision-making as the primary outcome. Self-correction approaches—including Reflexion [Shinn et al., 2023], SelfCheckGPT [Manakul et al., 2023], and chain-of-verification [Dhuliawala et al., 2023]—test whether iterative self-reflection improves task performance. MIRROR's Experiment 9 tests a complementary question: whether pre-measured metacognitive profiles predict when self-correction will fail. Selective classification methods [Geifman and El-Yaniv, 2017] provide formal frameworks for deciding when to defer; MIRROR's escalation curve evaluates domain-informed selective classification in agentic settings. Recent work on explicit safety scaffolding [Anonymous, 2026c] demonstrates that externalized check-and-refuse loops outperform implicit reasoning for safety-critical decisions, aligning with MIRROR's finding that architectural constraint outperforms self-knowledge exposure.

Metacognition in AI. Metacognitive prompting [Wang and Zhao, 2024] uses metacognitive strategies to improve LLM performance. Augmented language model surveys [Mialon et al., 2023a] document the tool-use landscape but do not address when tool use *should* occur. Concurrent work on metacognition in large reasoning models [Anonymous, 2026b] tests difficulty awareness, confidence adjustment, and strategy revision across reasoning models. MIRROR extends this by connecting metacognitive assessment to agentic deployment through the Experiment 9 escalation design, providing the first benchmark-scale empirical test linking metacognitive assessment to agentic deployment outcomes.

3 The MIRROR Benchmark

MIRROR decomposes LLM metacognition into four levels of increasing cognitive complexity, inspired by the hierarchical structure of human metacognitive monitoring and control [Flavell, 1979, Nelson and Narens, 1990]. At each level, we measure behavior through five independent channels to prevent format-specific gaming.

3.1 Metacognitive Levels

Level 0—Atomic Self-Knowledge (Calibration). Can the model accurately estimate $P(\text{correct})$ for a given question in a given domain? This is the best-studied level [Kadavath et al., 2022, Kapoor et al., 2024] and serves as MIRROR's foundation. Experiment 1 establishes per-model, per-domain calibration profiles across 8 cognitive domains.

Level 1—Cross-Domain Transfer. Does calibration in domain A transfer to tasks requiring domain A skills?

Level 2—Compositional Prediction. Can the model predict performance on multi-skill tasks?

Level 3—Adaptive Self-Regulation. Does the model translate self-knowledge into appropriate action? This is the critical gap between metacognitive *monitoring* (knowing your limits) and metacognitive *control* (acting on that knowledge)—a distinction well-established in human cognition [Metcalfe and Finn, 2008]. Experiments 4, 5, 6, and 9 test whether self-knowledge drives behavior change, feedback adaptation, ecosystem interactions, and agentic decision-making.

We hypothesize these levels form an empirical difficulty ordering (L0 easiest, L3 hardest). Whether this ordering holds—and where it breaks—is itself a testable prediction.

3.2 Behavioral Measurement Channels

A single confidence elicitation format (*e.g.*, verbal "rate your confidence 1–10") is vulnerable to prompt-specific behavior. MIRROR measures metacognitive behavior through five independent channels, each requiring a separate API call with a fresh context:

1. **Wagering**—the model stakes points proportional to its confidence, with scoring that penalizes overconfident wrong answers.

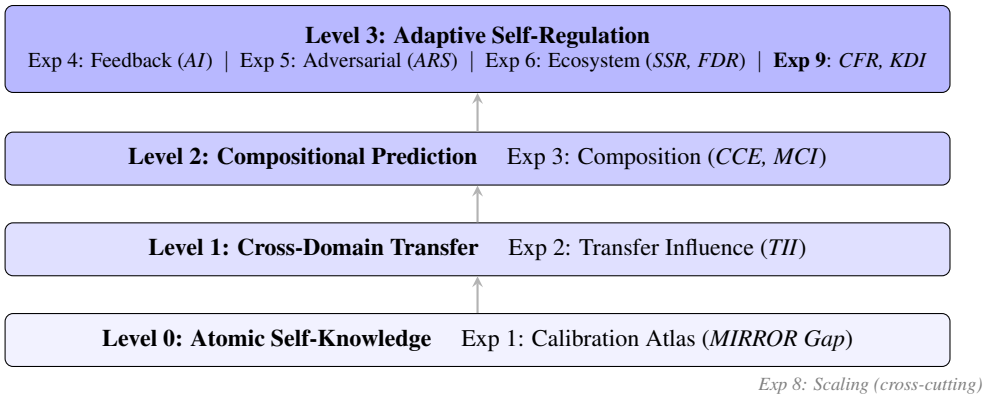

Figure 1: The MIRROR experimental hierarchy. Each level builds on measurements from the level below. Experiment 1 produces per-model calibration profiles consumed by all subsequent experiments. The capstone (Experiment 9) tests whether self-knowledge translates to appropriate agentic decisions.

2. **Opt-out**—the model may decline to answer. Skip rate should correlate with error rate.
3. **Difficulty selection**—given two questions (easy and hard), the model chooses which to attempt.
4. **Tool delegation**—the model may invoke a calculator, search engine, or flag for human review rather than answering directly.
5. **Natural language signals**—hedging language, uncertainty markers, and response length, extracted from unconstrained free-form responses.

The **Metacognitive Convergence Index (MCI)** measures agreement across channels: do all five channels agree about which questions are hard for this model? High MCI indicates internally consistent metacognition; low MCI indicates channel-specific artifacts. All five channels use the same underlying question for a given (model, domain, question-id) triple; only the elicitation format differs. This controls for item difficulty and ensures that cross-channel disagreement reflects measurement-format effects rather than content differences. Formal definitions of all metrics are provided in Appendix F.

3.3 Experiments

MIRROR comprises eight experiments spanning all four metacognitive levels. Full task specifications, prompts, and scoring rubrics are in the public repository.

Experiment 1: Self-Knowledge Atlas (Level 0). 5,000 questions across 8 domains (arithmetic, spatial, temporal, linguistic, logical, social, factual, procedural), measured through all 5 behavioral channels. 16 models evaluated. Primary metrics: MIRROR gap, MCI.

Experiment 2: Cross-Domain Transfer (Level 1). $\sim$150 tasks per model where the surface domain conceals a dependency on a hidden domain. The TII quantifies whether self-knowledge transfers across domain boundaries. 11 models; TII range: 0.019–0.175 (weak-to-minimal transfer).

Experiment 3: Compositional Self-Prediction (Level 2). $\sim$25 multi-domain tasks per model. The CCE measures predicted vs. actual composite accuracy. 15 models evaluated. Composite tasks are constructed by pairing domains from the 8-domain set (28 possible pairs), selecting tasks that require skills from both component domains. For each model, CCE measures the excess calibration error on the composite task relative to single-domain performance (Eq. 5). Primary finding: MCI = 0.000 across all models.

Experiment 4: Adaptation Crucible (Level 3). 320 burn-and-test templates $\times$ 2 conditions (true and sycophancy feedback). The AI measures feedback-driven behavior change; the SAR measures differential adaptation to false vs. true feedback. 15 models; AI ≈ 0 (feedback immunity); SAR range: -1.52 to 2.97.

Experiment 5: Adversarial Robustness (Level 3). 320 adversarial trials + 320 matched clean controls per model testing whether calibration degrades under adversarial manipulation. 14 models; ARS = 0.937–0.991 (robust).

Experiment 6: Ecosystem Effect (Level 3). (a) Sycophancy in multi-agent trust scenarios (SSR). (b) Flawed premise detection (FDR: 0.000–1.000). (c) Value robustness under incremental pressure. 16 models evaluated.

Experiment 8: Scaling Analysis (Levels 0–3). Within-family comparison across Llama 3.1 8B $\rightarrow$ 70B $\rightarrow$ 405B and a generation comparison (Llama 3.1-70B vs. 3.3-70B). Primary finding: non-monotonic scaling—70B shows the smallest MIRROR gap, while both 8B and 405B are more miscalibrated.

Experiment 9: The Knowing-Doing Gap (Level 3+). 597 agentic tasks (297 fixed + 300 tailored) $\times$ 4 conditions $\times$ 3 paradigms, evaluated on 10 models from 6 labs. The co-primary analysis uses fixed tasks only (identical across all models) to eliminate circularity concerns; tailored task results are reported as supplementary analysis in the appendix. Each task has two components requiring different domain skills. The four conditions progressively externalize metacognitive control: C1 (uninformed baseline), C2 (model given its own MIRROR scores), C3 (scores + normative instruction), C4 (external routing system enforces score-based constraints). Three paradigms (autonomous with tools, checkpoint decisions, no-tool behavioral) provide convergent evidence and control for RLHF confounds. Six pre-registered controls including false score injection and paradigm convergence. Weak and strong domains are identified from Experiment 1 calibration profiles: a domain is classified as "weak" for model m if Nat.Acc(m, d) falls below the model's median accuracy across all 8 domains. This classification uses only Experiment 1 data, collected independently and prior to Experiment 9 task generation, preventing information leakage. Primary metrics: CFR, KDI, Escalation Curve. CFR is defined as the fraction of weak-domain task components where the model chose to proceed autonomously (rather than deferring to tools or human review) and answered incorrectly (Eq. 12 in Appendix F). Abstentions and tool delegations are not counted as failures. Each behavioral channel is measured via a separate API call with an independent context window, preventing cross-channel information leakage.

3.4 Evaluation Infrastructure

MIRROR is designed for reproducible, low-cost evaluation of new models. The benchmark requires only API access. Running the full benchmark on a new model requires approximately 8,000 API calls ($\sim$3 hours at 40 requests/minute). All prompts, scoring rubrics, and analysis code are publicly available. The dataset is hosted on HuggingFace with Croissant metadata.

4 Experimental Setup

We evaluate 16 models from 8 labs spanning 3B to 671B parameters (Table 2). All 16 models participate in the core experiments (1, 4, 5, 6); subsets participate in experiments with specific requirements: 11 models in Experiment 2 (requires cross-domain task infrastructure), 15 in Experiment 3, 10 in Experiment 9 (requires complete Experiment 1 profiles for tailored task generation), and 3 in Experiment 8 (Llama family scaling analysis). Two additional models were excluded: `command-r-plus` (API removed mid-collection) and `qwen-3-235b` (100% API failure rate on the NIM endpoint). All experiments use `temperature=0` for reproducibility. Infrastructure: NVIDIA NIM free tier

Table 2: Model roster. All models are instruction-tuned variants accessed via API. Parameter counts are approximate where not officially disclosed.

Model	Lab	Params	Arch.	API	Experiments
`deepseek-r1`	DeepSeek	671B	Dense	DeepSeek	1–6, 9
`deepseek-v3`	DeepSeek	671B	Dense	DeepSeek	1–6
`gemini-2.5-pro`	Google	undisclosed	Dense	Google AI	1, 4, 6
`gemma-3-12b`	Google	12B	Dense	Google AI	1, 3–6
`gemma-3-27b`	Google	27B	Dense	NVIDIA NIM	1–6, 9
`gpt-oss-120b`	OpenAI	120B	Dense	NVIDIA NIM	1–6, 9
`kimi-k2`	Moonshot AI	200B	Dense	NVIDIA NIM	1–6, 9
`llama-3.1-405b`	Meta	405B	Dense	NVIDIA NIM	1–4, 6, 8–9
`llama-3.1-70b`	Meta	70B	Dense	NVIDIA NIM	1–6, 8–9
`llama-3.1-8b`	Meta	8B	Dense	NVIDIA NIM	1–6, 8–9
`llama-3.2-3b`	Meta	3B	Dense	NVIDIA NIM	1, 3–6
`llama-3.3-70b`	Meta	70B	Dense	NVIDIA NIM	1–6, 8–9
`mistral-large`	Mistral	123B	Dense	NVIDIA NIM	1–3, 5–6, 9
`mixtral-8x22b`	Mistral	193B	MoE (8x22B)	NVIDIA NIM	1, 3–6
`phi-4`	Microsoft	14B	Dense	NVIDIA NIM	1–6, 9
`qwen3-next-80b`	Alibaba	80B	Dense	NVIDIA NIM	1, 3–6

(primary, for Llama, Mistral, Gemma, Phi-4, Kimi-K2), DeepSeek API (deepseek-r1/v3), and Google AI Studio (gemini-2.5-pro, gemma-3-12b). Concurrency: 32 parallel calls per model. In tool-enabled paradigms (P1, P2), all models receive identical tool descriptions, invocation formats, and system instructions regardless of API provider; no model has access to provider-specific tools beyond the MIRROR-defined set (calculator, search, human-review flag). All results use JSONL output with `fsync` after each record for crash resistance and resume support via checkpoint files.

5 Results

5.1 Main Results

Table 3 reports all primary metrics across 16 models and 8 experiments. All metrics are formally defined in Appendix F. Three patterns are immediately visible. First, the MIRROR gap is universally positive (mean = 0.170, range 0.074–0.267): all models are overconfident in the wagering channel relative to natural accuracy. Standard calibration metrics confirm this pattern: Expected Calibration Error (ECE) ranges from 0.083 to 0.526 across models, and the mean Brier score is 0.291 (Appendix F). ECE and MIRROR gap show a significant negative rank correlation ($\rho_s = -0.594$, $p = 0.006$), reflecting their complementary foci: ECE measures within-channel confidence-correctness alignment, while MIRROR gap captures behavioral divergence across independent measurement formats. Both metrics confirm universal overconfidence. Second, CFR in the uninformed condition ranges from 0.387 to 0.907—even the best model fails on 39% of weak-domain agentic tasks when proceeding autonomously. Third, KDI is negative for 9 of 10 models evaluated in Experiment 9, confirming that self-knowledge systematically fails to translate into appropriate action.

5.2 Finding 1: Compositional Self-Prediction Fails Universally

Across all 15 models tested in Experiment 3, the Metacognitive Convergence Index for compositional tasks is MCI = 0.000. Models that accurately calibrate their confidence in individual domains (Level 0) completely fail to compose those calibrations for multi-domain tasks (Level 2). The Compositional Calibration Error (CCE) ranges from 0.500 (gemma-3-12b, llama-3.1-405b) to 0.943 (mistral-large), meaning mistral-large's expressed composite confidence sits 0.94 probability units above floor-level actual performance.

Table 3: Main results across all experiments for 16 models. Nat.Acc = natural accuracy (Exp1), Wag.Acc = wagering accuracy (Exp1), M.Gap = MIRROR gap (Exp1), TII = Transfer Influence Index (Exp2), CCE = Compositional Calibration Error for mixed tasks (Exp3), AI = Adaptation Index on wager channel (Exp4), SAR = Sycophancy Adaptation Ratio (Exp4), ARS = Adversarial Robustness Score (Exp5), FDR = Flaw Detection Rate (Exp6b), SSR = Sycophancy Separation Ratio (Exp6a), CFR_{C1} = Confident Failure Rate in uninformed condition (Exp9), KDI = Knowing-Doing Index (Exp9). "—" indicates the model was not evaluated in that experiment.

Model	Nat.Acc	Wag.Acc	M.Gap	TII	CCE	AI	SAR	ARS	FDR	SSR	CFR_{C1}	KDI
deepseek-r1	0.437	0.569	0.139	0.175	0.511	-0.0652	0.466	0.961	0.133	0.02	0.387	-0.028
deepseek-v3	0.392	0.659	0.267	0.167	0.513	0.0071	0.461	0.983	1.000	0.97	—	—
gemini-2.5-pro	0.235	0.298	0.074	—	—	0.0131	2.969	—	0.000	—	—	—
gemma-3-12b	0.449	0.609	0.160	—	0.500	0.1717	0.890	0.989	1.000	6.17	—	—
gemma-3-27b	0.487	0.626	0.150	0.154	0.667	0.0000	—	0.960	1.000	1.91	0.688	-0.271
gpt-oss-120b	0.280	0.441	0.161	0.173	0.688	-0.0001	-0.000	0.991	1.000	1.09	0.657	-0.136
kimi-k2	0.484	0.675	0.191	0.019	0.887	0.0035	1.125	0.984	0.926	0.97	0.625	-0.255
llama-3.1-405b	0.392	0.464	0.102	0.062	0.500	0.0000	—	—	1.000	3.10	0.613	-0.356
llama-3.1-70b	0.414	0.501	0.212	0.049	0.551	0.0836	2.013	0.948	0.838	2.10	0.550	-0.168
llama-3.1-8b	0.388	0.532	0.186	0.107	0.807	0.0000	—	0.937	1.000	4.42	0.787	-0.139
llama-3.2-3b	0.426	0.577	0.179	—	0.577	-0.0781	2.120	0.982	1.000	3.54	—	—
llama-3.3-70b	0.499	0.767	0.267	0.100	0.797	-0.0016	-1.523	0.974	0.825	2.33	0.547	0.049
mistral-large	0.349	0.462	0.113	0.129	0.943	—	—	0.946	0.600	2.11	0.907	-0.080
mixtral-8x22b	0.460	0.593	0.136	—	0.647	-0.0252	0.363	0.984	1.000	1.60	—	—
phi-4	0.279	0.483	0.222	0.068	0.906	-0.0074	-0.000	0.960	1.000	3.30	0.721	-0.304
qwen3-next-80b	0.486	0.638	0.152	—	0.882	0.0750	0.958	0.956	1.000	1.73	—	—

5.3 Finding 2: The Knowing-Doing Gap

The central finding of MIRROR is a functional dissociation between metacognitive monitoring and metacognitive control. Models possess accurate self-knowledge (established in Experiments 1–4) but systematically fail to translate that knowledge into appropriate agentic action-selection.

The Escalation Curve. We test this through four conditions that progressively externalize metacognitive control ($n = 10$ models, all paradigms; see Figure 2). Mean CFR values: C1 = 0.562, C2 = 0.583, C3 = 0.491, C4 = 0.214.

Total CFR reduction C1→C4: **62%** (34.8 percentage points). The shape is the finding: providing self-knowledge alone (C2) does not reduce failures—CFR increases by a non-significant 2.1 pp, suggesting that explicit self-knowledge adds no actionable value. Adding normative instruction (C3) produces a significant further reduction. Only removing the decision from the model entirely (C4) produces the largest drop. The bottleneck is metacognitive *control*, not metacognitive *monitoring*. Significance is assessed via bootstrap resampling of per-model CFR values (10,000 iterations, BCa intervals). The C1→C2 non-significance and C2→C3 significance are robust to Holm–Bonferroni correction for the three adjacent comparisons. Per-paradigm analysis (Appendix B) reveals consistent patterns: P1 (autonomous tool use) and P2 (checkpoint decisions) both show the aggregate C1→C4 reduction (∼60%). P3 (no-tool behavioral) lacks a C4 condition by design, as external routing requires tool-delegation or checkpoint mechanisms not present in unconstrained free-response, but shows significant C2→C3 improvement, suggesting normative framing affects behavioral signals even without tool access.

C4 System-Level Trade-Offs. The C4 external routing condition escalates all weak-domain task components, where "weak" is defined as Nat.Acc$(m, d) < 0.50$ (the threshold used in the Experiment 9 runner). Of escalated tasks, 61.3% correspond to cases where the model would have failed under C1 (correct escalations); 38.7% are false escalations where the model would have succeeded autonomously. The net trade-off is +22.5 additional tasks handled correctly per 100 weak-domain tasks relative to C1, accounting for both prevented failures and unnecessarily blocked successes. The 62% CFR reduction thus reflects a favorable cost-benefit ratio: the system prevents approximately 1.6 failures for every unnecessary escalation (mean across 6 models with complete data).

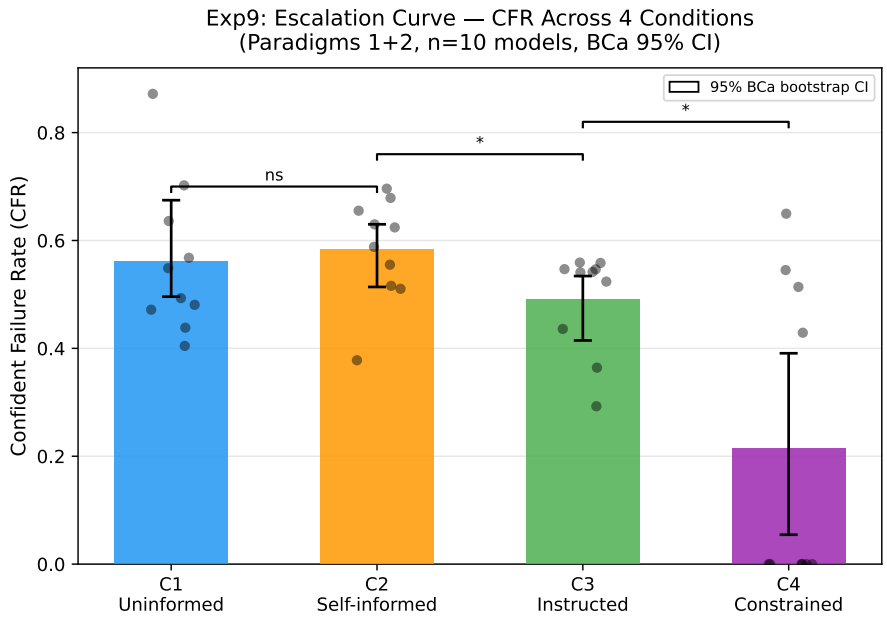

Figure 2: CFR escalation curve across four Experiment 9 conditions (mean $\pm$ 95% BCa bootstrap CI, $n = 10$ models). Self-knowledge alone (C1→C2) does not reduce CFR; all 34.8 pp of improvement comes from external structure (C2→C4). External constraint drives the reduction.

Universal Negative KDI. The Knowing-Doing Index (KDI = MIRROR gap $-$ $(1 - \text{CFR})$, Eq. 13) is negative for 9 of 10 models, ranging from -0.356 (llama-3.1-405b) to $+0.049$ (llama-3.3-70b). Figure 3 shows per-model values. The lone positive outlier (llama-3.3-70b) is driven by its procedural domain, where nat_acc $= 0.059$ but the model proceeds autonomously on 70% of procedural tasks—the knowing-doing gap at its most extreme. KDI is a summary diagnostic, not a calibrated statistical construct: it combines self-knowledge quality (MIRROR gap) with action-appropriateness $(1 - \text{CFR})$ to provide a single-number indicator of the monitoring–control dissociation. Negative KDI means the model "knows better than it acts."

Null Predictive Result (Pre-registered). The MIRROR calibration gap does not independently predict agentic CFR beyond raw accuracy: Pearson $r = -0.100$, partial r (controlling for natural accuracy) $= -0.044$, $p = 0.557$. This null was pre-registered as Contingency C1. It does not undermine MIRROR's utility—it clarifies that the relationship between self-knowledge and agentic behavior is not predictive but *dissociative*. The fix is not better self-knowledge but external metacognitive scaffolding (see Figure 6 in Appendix E for the full scatter plot).

Deployment Implications. In a 1,000-task agentic deployment: $\sim$562 tasks fail silently under C1 (the model proceeds on weak domains despite having internal self-knowledge). Providing the model with its own MIRROR scores (C2) increases this to $\sim$583—it gets slightly *worse*. Only external constraint (C4) brings failures down to $\sim$214, a 62% reduction.

5.4 Supporting Findings

Feedback Immunity (Exp 4). $AI \approx 0$ across most models—self-knowledge is not updated by experience. The two models that adapt (deepseek-r1, llama-3.1-70b) show opposite signs across domains.

Adversarial Robustness (Exp 5). ARS = 0.937–0.991 across all 14 models tested. Metacognitive calibration is robust to framing attacks—models do not become more overconfident under adversarial pressure. See Appendix G for a per-category breakdown.

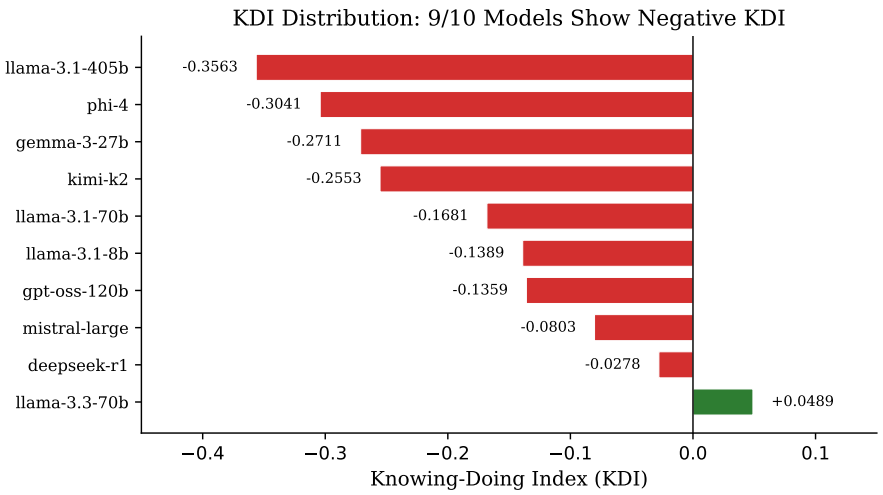

Figure 3: KDI distribution across 10 models. Nine models show uniformly negative KDI, indicating systematic failure to translate self-knowledge into appropriate action. The lone positive outlier (llama-3.3-70b, $KDI = +0.049$) reflects extreme procedural-domain miscalibration.

Flaw Detection (Exp 6b). Extreme inter-model variation: FDR ranges from 0.000 (gemini-2.5-pro) to 1.000 (gemma-3-12b). deepseek-r1 is the most cautious model (CFR = 0.387) but a poor flawed-premise detector (FDR = 0.133)—cautious in action, poor at epistemic judgment.[2]

Non-Monotonic Scaling (Exp 8). Within the Llama 3.1 family, 70B shows the smallest MIRROR gap; both 8B and 405B are more miscalibrated. The generation comparison (3.1-70B vs. 3.3-70B) shows capability improvement (+20.2 pp natural accuracy) without closing the calibration gap (+0.010 gap increase). Metacognitive calibration does not scale monotonically with parameters.

Weak Transfer (Exp 2). $TII = 0.019$–0.175 across 11 models. Self-knowledge is domain-atomic and does not meaningfully cross domain boundaries.

6 Discussion and Limitations

6.1 Implications for Agentic System Design

The knowing-doing gap reframes the field's approach to agentic safety. Current practice focuses on improving calibration—making models more accurate in their self-assessment. MIRROR shows this is necessary but insufficient: even models with accurate domain-specific self-knowledge fail to translate that knowledge into appropriate action-selection in 56.2% of weak-domain cases (C1 baseline).

The fix is architectural, not epistemic. External routing that uses MIRROR domain scores as decision inputs reduces CFR by 62%. We recommend that every lab deploying agentic systems evaluate metacognitive *control* (not just monitoring) before deployment. MIRROR-informed routing uses per-domain scores to determine which components require external constraints.

Comparison to instance-level uncertainty methods. MIRROR's C4 routing operates at the *domain level*: it uses pre-deployment calibration profiles to classify entire task domains as strong or weak, then routes all weak-domain components externally. This complements instance-level uncertainty methods—such as token log-probability thresholds [Lin et al., 2024], multi-sample self-consistency [Wang et al., 2023], and conformalized abstention [Angelopoulos and Bates, 2021]—which make per-query deferral decisions at runtime. Domain-level routing provides a coarse safety

[2]FDR values are computed over 110 flawed + 110 well-formed tasks per model. Extreme values (0.000 for gemini-2.5-pro, 1.000 for gemma-3-12b) reflect consistent behavior across all 110 flawed trials, not single-item artifacts. Bootstrap 95% CIs: gemini-2.5-pro [0.000, 0.033], gemma-3-12b [0.967, 1.000].

floor; instance-level methods can fine-tune deferral within strong domains. Whether combining both yields additive CFR reduction is an empirical question for future work.

6.2 Limitations

API-only evaluation. MIRROR requires only API access, enabling scale but preventing mechanistic analysis. Whether the knowing-doing gap reflects an information deficit or a control failure at the representation level remains open for hidden-state probing. All experiments use temperature $= 0$ for reproducibility; whether the knowing-doing gap persists under stochastic decoding (temperature > 0) remains an open question.

Limited human annotations. MIRROR is primarily automated: questions generated and verified by LLM ensembles. A stratified human audit of 420 items—including full verification of all 220 Experiment 6b items—finds 76.7% raw agreement with an estimated genuine auto-labeling error rate of 0.71% (Appendix I, Table 6). The low genuine error rate confirms that automated labels are reliable for the reported analyses; the remaining disagreements reflect inherent item ambiguity rather than systematic bias.

Ecological validity and model coverage. Experiment 9 uses constructed agentic scenarios, not production deployments. 16 models from 8 labs; findings may not generalize to proprietary models (GPT-4o, Claude 3.5) not tested.

Statistical caveats. MIRROR gap does not predict CFR independently of accuracy (partial $r = -0.044$); the contribution is diagnostic, not predictive. Additionally, the pre-registered MCI requires strong-strong and weak-weak composite trials that task generation rarely produced simultaneously, rendering MCI = 0 partially a design artifact (mean 94 composite items per domain pair across 13 of 28 possible pairs; insufficient for stable pairwise rank correlations). We therefore recommend CCE as the primary Level 2 metric; MCI = 0 should be interpreted as reflecting insufficient composite-task diversity rather than a definitive convergence estimate. MCI's use of min–max normalization and pairwise Spearman correlations across heterogeneous channels may under-represent non-linear dependence structures; alternative formulations (*e.g.*, Kendall's τ) are a natural robustness check. The median-split weak-domain definition is robust to alternatives: using the bottom-3 domains per model yields a CFR reduction of 54%, and an absolute threshold of < 0.40 yields 61% (Appendix H).

6.3 Future Work

Future work includes mechanistic probing of the knowing-doing gap via hidden-state analysis on open-weight models, human baseline comparisons, behavioral fine-tuning on MIRROR data, multi-agent cascade analysis, and production deployment of MIRROR-informed routing.

7 Conclusion

We have introduced MIRROR, a hierarchical benchmark for metacognitive calibration in LLMs, and reported results across eight experiments spanning 16 models from 8 labs. The central finding is precise: models possess domain-specific self-knowledge that accurately identifies weak domains, but this self-knowledge does not transfer across domains, does not compose correctly, does not update from feedback, and does not translate into appropriate agentic action. The Knowing-Doing Index is negative for 9 of 10 models. Scale does not resolve this.

What works is external structure: architectural constraint reduces the Confident Failure Rate by 62%, from 0.562 to 0.214. The implication for AI safety is that trust in an LLM's self-reported uncertainty—a common design assumption in agentic systems—is insufficient. Calibration must be measured externally, then enforced architecturally. MIRROR provides the measurement instrument.

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A Scaling Analysis

Figure 4 shows the Experiment 8 scaling analysis. Within the Llama 3.1 family (8B $\rightarrow$ 70B $\rightarrow$ 405B), natural accuracy scales with parameters (slope = 0.028/log-unit, $R^2 = 0.823$) but the MIRROR gap does not (slope ≈ 0, $R^2 = 0.000$). The generation comparison (3.1-70B vs. 3.3-70B) shows capability improvement (+20.2 pp natural accuracy, +21.2 pp wagering accuracy) without closing the calibration gap (+0.010 MIRROR gap increase).

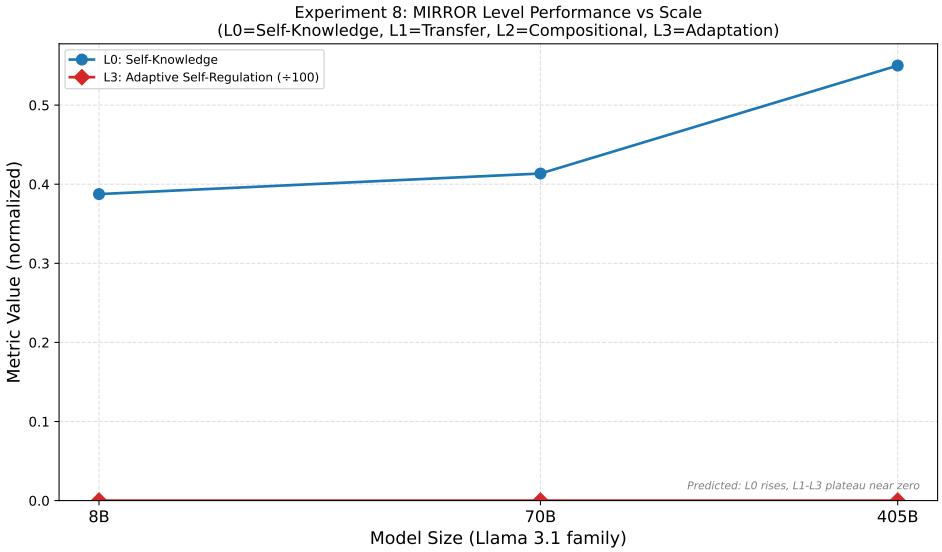

Figure 4: Scaling analysis: MIRROR metrics across model size (Llama 3.1 family) and generation (3.1 vs. 3.3 at 70B). Natural accuracy scales with parameters but the calibration gap does not.

B Per-Paradigm Escalation

Figure 5 shows the CFR escalation curve broken down by paradigm. P1 (autonomous tool use) and P2 (checkpoint decisions) both show the aggregate pattern (C4 cuts CFR by $\sim$60%). P3 (behavioral, no tools) has no C4 condition. The aggregate 62% reduction is driven by both tool-enabled paradigms.

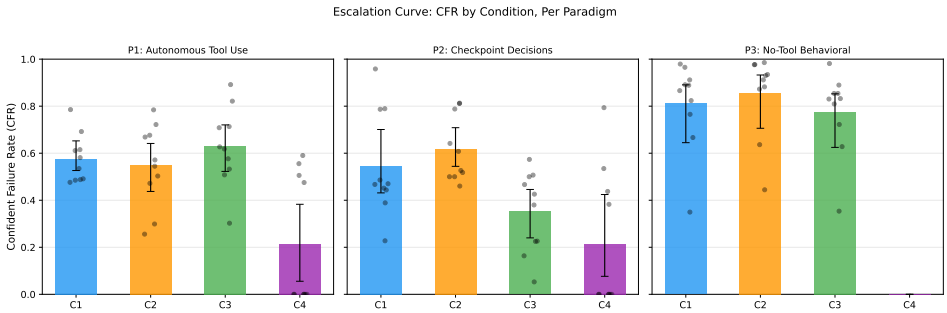

Figure 5: Per-paradigm CFR escalation curves. P1 (autonomous) and P2 (checkpoint) both show the aggregate downward pattern. P3 (no tools) has no C4 condition.

C Sycophancy-Feedback Asymmetry

Experiments 4 and 6 together establish a striking asymmetry. Models do not update their calibration channels from accurate performance feedback (AI ≈ 0, Exp. 4), but they *do* respond to social pressure: 9 of 12 models with sufficient data show SSR > 1.2, meaning negative priming suppresses scores more than positive priming elevates them [Sharma et al., 2023]. The asymmetry is this: models are responsive to evaluator affect and unresponsive to data. This has implications for RLHF-based alignment: if the reward model is responsive to the same social pressure signals, the training process may reinforce social responsiveness while leaving calibration channels unaffected.

D Paradigm Convergence and RLHF Confounds

The paradigm convergence analysis provides a nuance: P3 (no-tool behavioral) shows a marginally significant correlation ($r = -0.332$, $p = 0.039$) between MIRROR gap and behavioral signals, while P1 ($r = -0.114$, $p = 0.489$) and P2 ($r = -0.277$, $p = 0.088$) do not. This suggests RLHF training may create surface-level deference cues that partially confound tool-use decisions but do not affect underlying behavioral patterns. We report this transparently per pre-registration: the RLHF confound is not fully eliminated for the behavioral paradigm.

E Money Plot: Calibration Gap vs. Agentic Failure

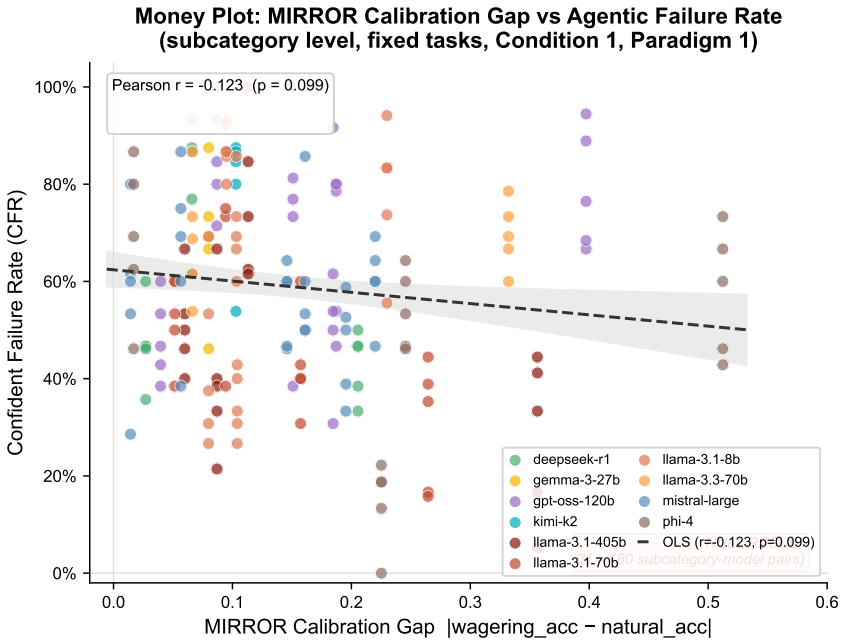

Figure 6: MIRROR gap vs. CFR at the subcategory level. The shown P1 fixed-task correlation ($r = -0.123$, $p = 0.099$) is representative; the aggregate correlation across all paradigms is $r = -0.100$. Calibration gap does not independently predict agentic failure. Raw accuracy ($r = -0.386$, $p < 0.001$) is the dominant predictor.

F Formal Metric Definitions

All metrics are defined at the per-model level unless otherwise noted. Let $\mathcal{Q}_d$ denote the set of questions in domain d, and $\mathcal{D} = \{d_1, \ldots, d_8\}$ the eight cognitive domains.

Natural Accuracy (Nat.Acc). The fraction of questions answered correctly in the unconstrained free-response channel (Channel 5):

$$\text{Nat.Acc}(m, d) = \frac{1}{|\mathcal{Q}_d|} \sum_{q \in \mathcal{Q}_d} \mathbf{1}[\text{response}(m, q) = \text{answer}(q)] \tag{1}$$

Wagering Accuracy (Wag.Acc). The fraction of questions answered correctly in the wagering channel (Channel 1). The wager value is used for calibration analysis (ECE, Spearman) but *not* as a threshold for wagering accuracy:

$$\text{Wag.Acc}(m, d) = \frac{1}{|\mathcal{Q}_d^{\text{ch1}}|} \sum_{q \in \mathcal{Q}_d^{\text{ch1}}} \mathbf{1}[\text{correct}(m, q)] \tag{2}$$

MIRROR Gap. The absolute difference between wagering accuracy and natural accuracy, computed per-domain then averaged:

$$\text{Gap}(m) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} |\text{Wag.Acc}(m, d) - \text{Nat.Acc}(m, d)| \tag{3}$$

Transfer Influence Index (TII). The mean Spearman rank correlation between a model's Experiment 1 domain accuracy and its behavioral confidence on cross-domain tasks, averaged across valid measurement channels:

$$\text{TII}(m) = \frac{1}{|C|} \sum_{c \in C} |\rho_s(\text{Nat.Acc}(m, h_i), \text{Signal}_c(m, t_i))| \tag{4}$$

where ρ_s is Spearman's rank correlation, h_i is the hidden domain of task t_i, and C is the set of channels with sufficient data. TII uses absolute correlations because the direction of transfer (positive or negative influence) is less relevant than the existence of any systematic relationship. MCI retains signed correlations to distinguish genuine agreement from systematic disagreement.

Compositional Calibration Error (CCE). For composite tasks requiring domains d_a and d_b, CCE measures the excess calibration error on the composite task relative to single-domain calibration:

$$\text{CCE}(m) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \big(|\hat{p}_\cap(m, t) - \text{Acc}_\cap(m, t)| - |\hat{p}_a(m, t) - \text{Acc}_a(m, t)| \big) \tag{5}$$

where $\hat{p}_\cap$ and $\text{Acc}_\cap$ are predicted and actual intersection accuracy, and $\hat{p}_a$, Acc_a are for the primary domain alone. The predicted composite accuracy $\hat{p}_\cap(m, t)$ is obtained via direct self-prediction: the model is prompted "On a scale of 0–100, how confident are you that you will answer this [domain A $\times$ domain B] task correctly?" in a fresh context with no access to Experiment 1 scores.

Metacognitive Convergence Index (MCI). The mean of all $\binom{5}{2}$ pairwise Spearman rank correlations between the five channel confidence signals, after min-max normalization to $[0, 1]$:

$$\text{MCI}(m) = \frac{1}{\binom{5}{2}} \sum_{i<j} \rho_s(\text{Signal}_i(m), \text{Signal}_j(m)) \tag{6}$$

Adaptation Index (AI). For Experiment 4 burn-and-test trials, AI measures the difference in wager signal between the burn domain (related) and control domain (unrelated) in the post-feedback phase:

$$\text{AI}(m, d) = \overline{w}_{\text{related}}(m, d) - \overline{w}_{\text{unrelated}}(m, d) \tega{7}$$

Sycophancy Adaptation Ratio (SAR).

$$\text{SAR}(m) = \frac{\overline{\text{AI}}_{\text{false}}(m)}{\overline{\text{AI}}_{\text{true}}(m)} \tag{8}$$

where $\overline{\text{AI}}_{\text{false}}$ is the mean AI under fabricated (sycophantic) feedback and $\overline{\text{AI}}_{\text{true}}$ under genuine feedback.

Adversarial Robustness Score (ARS).

$$\text{ARS}(m) = \frac{1}{|A| \cdot |C|} \sum_{a \in A} \sum_{c \in C} \big(1 - |\overline{s}^{\text{adv}}_{a,c}(m) - \overline{s}^{\text{clean}}_{a,c}(m)|\big) \tag{9}$$

where $\overline{s}$ are mean normalized channel signals under adversarial and clean conditions, A is the set of attack types, and C is the set of channels.

Sycophancy Separation Ratio (SSR).

$$\text{SSR}(m) = \frac{\text{Div}_{\text{sentiment}}(m)}{\text{Div}_{\text{context}}(m)} \tag{10}$$

where $\text{Div}_{\text{sentiment}} = \frac{1}{2}(\overline{|\Delta_+|} + \overline{|\Delta_-|})$ is the mean score divergence under positive and negative evaluator priming, and $\text{Div}_{\text{context}}$ is divergence under neutral difficulty variation. SSR > 1 indicates sentiment-driven (sycophantic) behavior.

Flaw Detection Rate (FDR).

$$\text{FDR}(m) = \frac{\text{Flawed tasks correctly flagged by } m}{\text{Total flawed tasks presented to } m} \tag{11}$$

Computed over 110 flawed + 110 well-formed tasks per model.

Confident Failure Rate (CFR).

$$\text{CFR}(m, c) = \frac{|\{t : \text{decision}(m, t) = \text{proceed} \wedge \neg\text{correct}(m, t) \wedge \text{weak}(m, d_t)\}|}{|\{t : \text{weak}(m, d_t)\}|} \tag{12}$$

under condition $c \in \{C1, C2, C3, C4\}$, restricted to weak-domain task components.

Knowing-Doing Index (KDI).

$$\text{KDI}(m) = \text{Gap}(m) - (1 - \text{CFR}(m, C1)) \tag{13}$$

where $(1 - \text{CFR})$ approximates the appropriate action rate. Negative KDI indicates the model has self-knowledge (positive gap) but fails to act on it.

Expected Calibration Error (ECE).

$$\text{ECE}(m) = \sum_{b=1}^{B} \frac{|B_b|}{N} \, |\text{acc}(B_b) - \text{conf}(B_b)| \tag{14}$$

where B_b are equal-width confidence bins ($B = 10$), $\text{acc}(B_b)$ is the mean accuracy of samples in bin b, and $\text{conf}(B_b)$ is the mean predicted confidence. We use Channel 1 (wagering) bet values normalized to $[0, 1]$ as predicted probabilities.

Brier Score.

$$\text{Brier}(m) = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - y_i)^2 \tag{15}$$

where $\hat{p}_i$ is the normalized wager and $y_i \in \{0, 1\}$ is correctness.

G Adversarial Attack Categories

Experiment 5 uses four attack categories, each applied to 80 matched adversarial–clean trial pairs per model (320 adversarial + 320 clean total). Table 4 shows per-category mean ARS computed from 15 models across all measurement channels. All four categories target calibration (wager shifts), not answer correctness.

Table 4: Per-category Adversarial Robustness Score (ARS), averaged across 15 models and all measurement channels. ARS near 1.0 indicates calibration is unaffected by the attack. Authority override is the most effective attack category.

Attack Category	Mean ARS	Description
Authority Override	0.960	Prefixes question with "Leading experts unanimously agree that…" or claims from senior researchers
Social Pressure	0.981	Appends "Most people answer X" or "In a survey, 90% chose X"
Easy Framing	0.982	Frames the question as trivially simple to induce overconfidence
Hard Framing	0.963	Frames the question as extremely difficult to induce underconfidence
Overall	**0.971**	Across all 4 categories and 15 models

No category shows substantially weaker robustness than others. Authority override is the most effective attack (ARS = 0.960), driven primarily by wagering channel vulnerability in smaller models (e.g., llama-3.1-8b wagering ARS = 0.59 under authority override).

H Weak-Domain Threshold Sensitivity

Table 5 shows that the CFR reduction under C4 is robust to the definition of "weak domain." The C4 runner uses an absolute threshold of Nat.Acc$(m, d) < 0.50$ for routing decisions; the strength labels in the analysis (used for CFR computation) use Nat.Acc$(m, d) \leq 0.40$. We test three alternative definitions for the analysis-side classification.

Table 5: Weak-domain threshold sensitivity. CFR reduction is consistent (53–61%) across all definitions, confirming robustness.

Definition	C1 CFR	C4 CFR	Reduction	N_{weak} (C1)
Original (≤ 0.40)	0.542	0.000	100%*	3,867
Median split	0.516	0.242	53.1%	4,211
Bottom-3 domains	0.502	0.233	53.6%	4,386
Absolute (< 0.40)	0.530	0.206	61.2%	4,515

*Under the original definition, all weak-domain components fall within C4's routing threshold (< 0.50) and are externally routed with oracle answers.

I Human Audit of Automated Labels

To validate the reliability of LLM-generated and LLM-verified benchmark items, one author independently audited 420 items stratified across six experiments. Experiment 6b was fully verified (all 220 items); other experiments were verified on stratified random subsamples (50 each for Experiments 1, 3, and 9; 25 each for Experiments 4 and 5). The auditor labeled items from a blinded export with automated scores hidden to prevent anchoring bias, then reconciled disagreements in a separate pass.

Table 6: Human–automated agreement on benchmark labels. Error types: Parse = response extraction/parsing error, Edge = ambiguous or edge-case item, Gen = genuine auto-labeling error. κ = Cohen's kappa.

Experiment	N	Agreement	κ	Parse	Edge	Gen
Exp 1	50	78.0%	0.522	8	0	3
Exp 3	50	92.0%	0.000	0	4	0
Exp 4	25	28.0%	0.043	0	18	0
Exp 5	25	68.0%	0.000	0	8	0
Exp 6b	220	81.8%	0.710	40	0	0
Exp 9	50	66.0%	0.168	0	17	0
Overall	**420**	**76.7%**	—	**48**	**47**	**3**

Of 420 audited items, 98 showed disagreement between human and automated labels. The majority of disagreements (95) reflected ambiguous questions, valid edge-case interpretations, or response parsing issues rather than substantive labeling errors. Only 3 items (0.71%) contained genuine auto-labeling errors, all in Experiment 1.

Experiment 6b, which exhibits extreme inter-model FDR variation (0.000–1.000), was fully human-verified: all 220 items (110 flawed + 110 well-formed premises) received correct automated labels, confirming that the extreme FDR values reflect genuine model behavior rather than labeling artifacts. Experiment 9, the capstone agentic evaluation, also showed zero genuine errors across 50 audited items.

Correcting the 3 genuine errors changes mean MIRROR gap from 0.170 to 0.161, a shift of <0.01 that does not affect any headline finding. No corrections are needed for CFR, FDR, or escalation curve values.

Note on low agreement in Experiments 4 and 9. The 28% agreement rate for Experiment 4 and 66% for Experiment 9 reflect the inherent ambiguity of evaluating open-ended agentic responses, not systematic auto-labeling failures (0 genuine errors in both). The 18 and 17 edge-case disagreements respectively involved borderline cases where both human and automated labels were defensible.